OpenSearch-VL: An Open Recipe for Frontier Multimodal Search Agents

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Research Questions

- How can we create an open-source training recipe for frontier multimodal search agents that combines high-quality training data, diverse tools, and robust reinforcement learning?
- How can we handle cascading tool failures in multi-turn RL training while preserving useful pre-failure reasoning?
- How can we construct training data that encourages multi-hop reasoning rather than single-step retrieval shortcuts in multimodal search tasks?

Core Idea

This paper addresses the reproducibility crisis in multimodal search agents - while commercial models excel at visual search and reasoning, their training recipes remain proprietary. The key insight is threefold: (1) construct training data from Wikipedia graphs with "fuzzy entity rewriting" to force multi-hop reasoning instead of shortcuts, (2) equip agents with visual enhancement tools (OCR, super-resolution, cropping) beyond just search, and (3) handle inevitable tool failures in RL training through "fatal-aware masking" that preserves good reasoning before failures while ignoring post-failure noise. The fatal-aware GRPO uses one-sided advantage clamping - if a trajectory fails but its pre-failure reasoning was above average, it gets positive reinforcement; if below average, it gets zero gradient instead of negative punishment.

Key Formula

One-sided Advantage Clamping

$$\hat{A}_i = \begin{cases} \tilde{r}_i & \text{if } f_i = L_i + 1 \text{ (non-fatal)}, \\ \max(\tilde{r}_i, 0) & \text{if } f_i \leq L_i \text{ (fatal)} \end{cases}$$

Where \hat{A}_i is the final advantage, \tilde{r}_i is the group-normalized reward, f_i is the fatal step index, and L_i is trajectory length. For fatal trajectories, negative advantages are clamped to zero to avoid penalizing valid pre-failure reasoning, while positive advantages are preserved to reinforce good partial solutions.

Fatal-aware Token Mask

$$M(y_{i,t}) = M_{gen}(y_{i,t}) \cdot \mathbb{1}[s(t) < f_i]$$

Where M(y_{i,t}) is the training mask, M_{gen}(y_{i,t}) indicates policy-generated tokens, s(t) maps token index to step index, and f_i is the fatal step. This masks both environmental tokens and all tokens after tool failure cascades, preventing noisy gradients from post-failure reasoning.

Key Figure

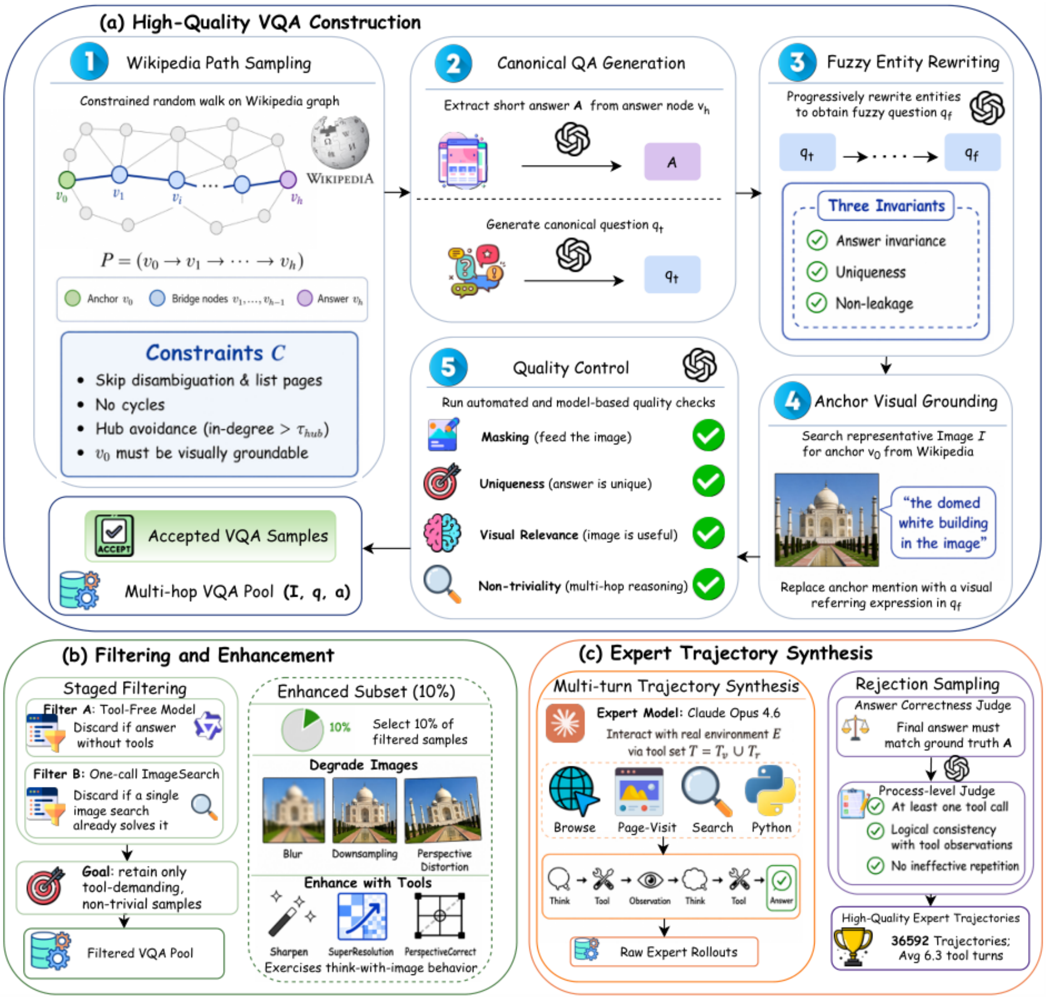

Figure 1 | **Overview of the data curation pipeline.** (a) Starting from the English Wikipedia hyperlink graph, we construct high-quality multi-hop VQA instances by sampling constrained paths, generating canonical question–answer pairs, rewriting them into fuzzy questions, grounding anchor entities with representative images, and applying automated quality control. (b) We then perform staged filtering to retain only tool-demanding, non-trivial samples, and create an enhanced subset through image degradation and tool-based restoration to encourage think-with-image behavior. (c) Finally, we synthesize multi-turn expert trajectories in a real tool environment and apply rejection sampling with answer-correctness and process-level judges, yielding the final high-quality trajectories used for the following stage training.

Figure 1: Shows the complete data curation pipeline from Wikipedia sampling to expert trajectory synthesis, illustrating the key innovations of fuzzy entity rewriting and source-anchor visual grounding that prevent single-hop shortcuts.

Limitations

- Relies on external APIs (Serper, PaddleX OCR) that introduce training instability and reproducibility challenges
- Uses proprietary GPT-4o judges for reward evaluation, which are costly and version-dependent
- Process reward models only score textual queries while ignoring visual operations like cropping
- Exact numerical reproducibility is challenging due to external API dependencies and prohibitive cost of multi-seed evaluations